# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 84.5 ± 1.36 |
| Correlation Stability | 87.5 ± 1.63 |
| Distribution Consistency | 86.0 ± 1.26 |
| Entropy Preservation | 95.9 ± 0.30 |
| Mean Z Score Drift | 0.1 ± 0.02 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 0.0 ± 0.00 |
| Data Density | 100.0 ± 0.00 |
| Coverage Score | 99.2 ± 1.25 |
| Feature Accuracy | 93.5 ± 1.57 |
| Novelty Score | 0.9 ± 0.01 |

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## Dataset Audit: breast_cancer_coimbra.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 116 |
| Columns | 10 |
| Numeric features | 10 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | gpt-4o-mini |
| Folds (*13) | 10 |
| Context % (*14) | 0% |
| Generation % (*15) | 100% |
| Join context (*16) | False |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| Age | 57.302 | 60.022 | +2.720 | 4.7% | OK |
| BMI | 27.582 | 27.425 | -0.157 | 0.6% | OK |
| Glucose | 97.793 | 97.065 | -0.728 | 0.7% | OK |
| Insulin | 10.012 | 9.842 | -0.170 | 1.7% | OK |
| HOMA | 2.695 | 2.839 | +0.144 | 5.3% | OK |
| Leptin | 26.615 | 25.780 | -0.835 | 3.1% | OK |
| Adiponectin | 10.181 | 10.413 | +0.232 | 2.3% | OK |
| Resistin | 14.726 | 17.582 | +2.856 | 19.4% | Warn |
| MCP.1 | 534.647 | 535.728 | +1.081 | 0.2% | OK |
| Classification | 0.552 | 0.587 | +0.035 | 6.4% | OK |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 84.5 ± 1.36 |
| Correlation Stability (*2) | 87.5 ± 1.63 |
| Distribution Consistency (*3) | 86.0 ± 1.26 |
| Entropy Preservation (*4) | 95.9 ± 0.30 |
| Mean Z Score Drift (*5) | 0.1 ± 0.02 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 0.0 ± 0.00 |
| Data Density (*8) | 100.0 ± 0.00 |
| Coverage Score (*9) | 99.2 ± 1.25 |

| | |
|---|---|
| Feature Accuracy (*10) | 93.5 ± 1.57 |
| Novelty Score (*11) | 0.9 ± 0.01 |

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.4958333333333333 | 0.4666666666666666 | 0.45 | -0.04583333333333333 | |
| F1 | 0.6547289527475286 | 0.5108795734142783 | 0.5160847655221057 | -0.13864418722542293 | |

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7458333333333333 | 0.5999999999999999 | 0.6833333333333333 | -0.06250000000000001 | |
| F1 | 0.754839950419102 | 0.6060295566502463 | 0.6764029480986002 | -0.07843700232050174 | |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7041666666666666 | 0.6083333333333334 | 0.6708333333333334 | -0.03333333333333326 | |
| F1 | 0.7282173803208286 | 0.6159828332586953 | 0.6844062925802057 | -0.04381108774062299 | |

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.4625 | 0.4833333333333334 | 0.5166666666666667 | 0.05416666666666675 | |
| F1 | 0.5361601301923884 | 0.5369781960716332 | 0.5901204630556689 | 0.05396033286328057 | |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5833333333333333 | 0.5833333333333334 | 0.5791666666666666 | -0.004166666666666674 | |
| F1 | 0.5965064710909768 | 0.5964056991571223 | 0.5754404883674661 | -0.021065982723510696 | |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7166666666666666 | 0.6583333333333332 | 0.6125 | -0.1041666666666667 | |
| F1 | 0.7136009412875979 | 0.6506874976327124 | 0.6143054042731462 | -0.09929553701445173 | |

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7416666666666666 | 0.5916666666666666 | 0.6791666666666667 | -0.0625 | |
| F1 | 0.7647514245014245 | 0.5755290104303754 | 0.6816657464918334 | -0.08308567800959107 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6416666666666667 | 0.6416666666666666 | 0.6208333333333333 | -0.02083333333333336 | |
| F1 | 0.5557236965932618 | 0.5733643940899291 | 0.4916140895717556 | -0.06410960702150634 | |

### Model: LR

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7083333333333334 | 0.6375 | 0.6666666666666666 | -0.04166666666666664 | |
| F1 | 0.7179811943844927 | 0.659584149422859 | 0.6761961371961372 | -0.04178505718835551 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.725 | 0.6 | 0.6791666666666666 | -0.04583333333333334 | |
| F1 | 0.7359528619528619 | 0.5966533466533466 | 0.6878969648265 | -0.04805589712636189 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 48.0 | 23.5 | 70.0 | 2.707 | 0.467408667 | 8.8071 | 9.7024 | 7.99585 | 417.114 | 0.0 |
| 2 | 83.0 | 20.69049454 | 92.0 | 3.115 | 0.706897333 | 8.8438 | 5.429285 | 4.06405 | 468.786 | 0.0 |
| 3 | 82.0 | 23.12467037 | 91.0 | 4.498 | 1.009651067 | 17.9393 | 22.43204 | 9.27715 | 554.697 | 0.0 |
| 4 | 68.0 | 21.36752137 | 77.0 | 3.226 | 0.612724933 | 9.8827 | 7.16956 | 12.766 | 928.22 | 0.0 |
| 5 | 86.0 | 21.11111111 | 92.0 | 3.549 | 0.8053864 | 6.6994 | 4.81924 | 10.57635 | 773.92 | 0.0 |
| 6 | 49.0 | 22.85445769 | 92.0 | 3.226 | 0.732086933 | 6.8317 | 13.67975 | 10.3176 | 530.41 | 0.0 |
| 7 | 89.0 | 22.7 | 77.0 | 4.69 | 0.890787333 | 6.964 | 5.589865 | 12.9361 | 1256.083 | 0.0 |
| 8 | 76.0 | 23.8 | 118.0 | 6.47 | 1.883201333 | 4.311 | 13.25132 | 5.1042 | 280.694 | 0.0 |
| 9 | 73.0 | 22.0 | 97.0 | 3.35 | 0.801543333 | 4.47 | 10.358725 | 6.28445 | 136.855 | 0.0 |
| 10 | 75.0 | 23.0 | 83.0 | 4.952 | 1.013839467 | 17.127 | 11.57899 | 7.0913 | 318.302 | 0.0 |
| 11 | 34.0 | 21.47 | 78.0 | 3.469 | 0.6674356 | 14.57 | 13.11 | 6.92 | 354.6 | 0.0 |
| 12 | 29.0 | 23.01 | 82.0 | 5.663 | 1.145436133 | 35.59 | 26.72 | 4.58 | 174.8 | 0.0 |
| 13 | 25.0 | 22.86 | 82.0 | 4.09 | 0.827270667 | 20.45 | 23.67 | 5.14 | 313.73 | 0.0 |
| 14 | 24.0 | 18.67 | 88.0 | 6.107 | 1.33 | 8.88 | 36.06 | 6.85 | 632.22 | 0.0 |
| 15 | 38.0 | 23.34 | 75.0 | 5.782 | 1.06967 | 15.26 | 17.95 | 9.35 | 165.02 | 0.0 |
| 16 | 44.0 | 20.76 | 86.0 | 7.553 | 1.6 | 14.09 | 20.32 | 7.64 | 63.61 | 0.0 |
| 17 | 47.0 | 22.03 | 84.0 | 2.869 | 0.59 | 26.65 | 38.04 | 3.32 | 191.72 | 0.0 |
| 18 | 61.0 | 32.03895937 | 85.0 | 18.077 | 3.790144333 | 30.7729 | 7.780255 | 13.68392 | 444.395 | 0.0 |
| 19 | 64.0 | 34.5297228 | 95.0 | 4.427 | 1.037393667 | 21.2117 | 5.46262 | 6.70188 | 252.449 | 0.0 |
| 20 | 32.0 | 36.51263743 | 87.0 | 14.026 | 3.0099796 | 49.3727 | 5.1 | 17.10223 | 588.46 | 0.0 |

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## B.1 Audit Trace: Fold 1 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 81.0 | 32.95923771008302 | 136.0 | 23.5732163602986 | 12.025995101396092 | 41.15557253796093 | 1.13388866104091 | 25.3748714672266 | 307.884358584751 | 1.0 |
| 2 | 81.0 | 20.88939751170805 | 108.0 | 6.53843001639052 | 1.70777002368061 | 17.05140323456426 | 8.01625949162406 | 18.4850212157915 | 851.839083342409 | 1.0 |
| 3 | 60.0 | 38.91916095144219 | 113.0 | 5.13439818803247 | 2.14525243230099 | 57.81964859933238 | 8.68264947961231 | 19.9504999578002 | 211.696747965185 | 1.0 |
| 4 | 85.0 | 26.18045582022927 | 88.0 | 9.17150255330911 | 2.53566873388072 | 9.96276214475709 | 13.0644662199241 | 24.5885106128880 | 654.071853683889 | 1.0 |
| 5 | 93.0 | 34.23218259428535 | 156.0 | -13.21476633021246 | 13.97553986674578 | 56.51468503202554 | 17.47488124755737 | 9.51347872625053 | 347.281614931798 | 1.0 |
| 6 | 56.0 | 22.95834022805897 | 85.0 | 15.89000782081880 | -0.8745112550144429 | 22.23587168003229 | 18.34492064631423 | 11.0638291917124 | 854.207609295668 | 0.0 |
| 7 | 64.0 | 22.22222222 | 98.0 | 5.7 | 1.37788 | 12.1905 | 4.783985 | 13.91245 | 395.976 | 1.0 |
| 8 | 57.0 | 34.83814777 | 95.0 | 12.548 | 2.940414667 | 33.1612 | 2.36495 | 9.9542 | 655.834 | 1.0 |
| 9 | 68.0 | 21.08281329 | 102.0 | 6.2 | 1.55992 | 9.6994 | 8.574655 | 13.74244 | 448.799 | 1.0 |
| 10 | 42.0 | 21.35991456 | 93.0 | 2.999 | 0.6879706 | 19.0826 | 8.462915 | 17.37615 | 321.919 | 1.0 |

## B.2 Audit Trace: Fold 2 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 59.0 | 32.27462979257319 | 97.0 | 32.20600313910005 | 3.926510657038697 | 39.69869803473557 | 24.56489793481770 | 15.80999552512381 | 698.782474610846 | 0.0 |
| 2 | 59.0 | 28.19996313901630 | 102.0 | -4.265549965234787 | -0.2527837628330212 | 13.84071644064400 | 7.798175879262571 | 14.85317219619962 | 368.697142248938 | 1.0 |
| 3 | 74.0 | 23.14608362528163 | 98.0 | 11.33998793975747 | 4.077621210894197 | 10.16896782456752 | 12.52156143872114 | 10.91981631129162 | 271.486182876874 | 1.0 |
| 4 | 64.0 | 28.82757867414464 | 67.0 | 14.26319095753717 | 5.549509022543837 | 11.51443137748339 | 7.984946299273596 | 10.98509643687945 | 144.170802394369 | 0.0 |
| 5 | 56.0 | 21.08697430161562 | 96.0 | 2.888556654493374 | 0.7006494509542796 | 20.15267257606840 | 7.313944607335163 | 24.94966065350128 | 778.565866482648 | 1.0 |
| 6 | 34.0 | 22.02885117845653 | 99.0 | 3.869080966120561 | 1.217518306944071 | 31.79148353926464 | 19.42313452944738 | 4.194298831213727 | 331.996123686744 | 0.0 |
| 7 | 50.0 | 38.57875854 | 106.0 | 6.703 | 1.752611067 | 46.6401 | 4.667645 | 11.78388 | 887.16 | 0.0 |
| 8 | 57.0 | 34.83814777 | 95.0 | 12.548 | 2.940414667 | 33.1612 | 2.36495 | 9.9542 | 655.834 | 1.0 |
| 9 | 38.0 | 22.4996371 | 95.0 | 5.261 | 1.232827667 | 8.438 | 4.77192 | 15.73606 | 199.055 | 1.0 |
| 10 | 81.0 | 31.64036818 | 100.0 | 9.669 | 2.38502 | 38.8066 | 10.636525 | 29.5583 | 426.175 | 1.0 |

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## B.3 Audit Trace: Fold 3 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 48.0 | 23.2884243970665 | 91.0 | 3.13671689598151 | 0.66014725087634 | 42.0317616200736 | 29.6665949558903 | 5.05325311500056 | 92.5606704661621 | 1.0 |
| 2 | 39.0 | 25.0102172586333 | 85.0 | 3.49507066372428 | 0.75148724321169 | 31.8175969821380 | 25.3019953490379 | 9.3470195112508 | 236.626392053100 | 1.0 |
| 3 | 68.0 | 21.9419014240366 | 140.0 | 25.1769958621005 | 0.22638566621037 | 49.1556941289782 | 6.98216371507028 | 35.6840205303838 | 1153.88572584266 | 1.0 |
| 4 | 58.0 | 35.4668154034983 | 73.0 | 13.8784868659428 | 2.45417112732916 | 34.3194547824640 | 4.37659772094066 | -7.57906148582148 | 2032.54042775924 | 1.0 |
| 5 | 58.0 | 24.7701980465614 | 80.0 | 10.6668973104680 | -1.99530668351203 | 18.7392501564925 | 41.3389881330941 | 5.10868549968302 | 214.771900613351 | 0.0 |
| 6 | 47.0 | 26.3331343105408 | 92.0 | 7.48105354622192 | 1.05450450217526 | 6.19010398599655 | 5.12955372029487 | 20.3415067173499 | 291.056791775834 | 0.0 |
| 7 | 43.0 | 31.25 | 103.0 | 4.328 | 1.099600533 | 25.7816 | 12.71896 | 38.6531 | 775.322 | 1.0 |
| 8 | 24.0 | 18.67 | 88.0 | 6.107 | 1.33 | 8.88 | 36.06 | 6.85 | 632.22 | 0.0 |
| 9 | 71.0 | 25.51020408 | 112.0 | 10.395 | 2.871792 | 19.0653 | 5.4861 | 42.7447 | 799.898 | 1.0 |
| 10 | 49.0 | 21.36752137 | 78.0 | 2.64 | 0.507936 | 6.3339 | 3.886145 | 22.94254 | 737.672 | 1.0 |

## B.4 Audit Trace: Fold 4 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 51.0 | 22.2148011114364 | 91.0 | 3.66763939582939 | 0.50755567417999 | -3.75500704273360 | 16.4569116483233 | 5.55111594632116 | 1030.72185393581 | 1.0 |
| 2 | 68.0 | 34.0046555262910 | 128.0 | 19.4695623055802 | 21.8976076017104 | 18.6444857248487 | 9.29100376828850 | 22.7809132976527 | 1114.00326988336 | 1.0 |
| 3 | 43.0 | 30.8981105147413 | 119.0 | 12.1330295607237 | 1.63431563184915 | 29.4317914213417 | 4.72986931401027 | 17.2786408643372 | 644.377645005271 | 1.0 |
| 4 | 71.0 | 20.0954633172711 | 82.0 | 7.76692967542712 | 3.09442999096123 | 20.6529868526968 | 12.2894818011036 | 14.4399158458512 | 611.574470083464 | 0.0 |
| 5 | 77.0 | 23.7895893786630 | 97.0 | 21.0271506179992 | -0.08697617284510 | 3.87508004085713 | 2.21003781209480 | -6.36037836823913 | -281.773033095764 | 0.0 |
| 6 | 65.0 | 21.6110125261251 | 98.0 | 0.80635734229182 | 0.93790376426044 | 11.3146214717938 | 5.54237233820389 | 14.0632955006440 | 362.008410585330 | 1.0 |
| 7 | 75.0 | 30.48 | 152.0 | 7.01 | 2.628282667 | 50.53 | 10.06 | 11.73 | 99.45 | 1.0 |
| 8 | 60.0 | 26.34929208 | 103.0 | 5.138 | 1.305394533 | 24.2998 | 2.19428 | 20.2535 | 378.996 | 0.0 |
| 9 | 77.0 | 35.58792924 | 76.0 | 3.881 | 0.727558133 | 21.7863 | 8.12555 | 17.2615 | 618.272 | 0.0 |
| 10 | 72.0 | 23.62 | 105.0 | 4.42 | 1.14478 | 21.78 | 17.86 | 4.82 | 195.94 | 1.0 |

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## B.5 Audit Trace: Fold 5 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 50.0 | 24.8619884166849 | 92.0 | 26.8787458497943 | 6.44165341901822 | 13.808682135049 | 17.0977252331268 | 17.0641474956020 | -114.380385785774 | 0.0 |
| | | 4 | | 06 | 15 | | 16 | 54 | 48 | |
| 2 | 46.0 | 33.7891733358288 | 90.0 | 6.30203800953486 | 2.68412815594148 | 75.7851440841607 | 7.11769731069790 | 6.68573438565499 | 310.643890620868 | 0.0 |
| | | 06 | | 2 | 22 | 3 | 9 | 4 | | |
| 3 | 73.0 | 31.5469617077813 | 156.0 | 19.0054133337699 | 7.01072460270665 | 129.358730931702 | 6.57355112201301 | 9.35355263053159 | 1362.15444385765 | 1.0 |
| | | 3 | | 62 | 9 | 84 | 1 | 9 | 11 | |
| 4 | 47.0 | 31.1626844254851 | 106.0 | 12.1270698683509 | 3.32127710012015 | 47.6258101828059 | 12.1127690771046 | 11.4490874172624 | 738.950813059018 | 0.0 |
| | | 18 | | 17 | 1 | 14 | 94 | 19 | | |
| 5 | 66.0 | 28.1720373790778 | 67.0 | 20.2624083912146 | 2.97735704757222 | -11.2014520488724 | 19.8857378353890 | 8.68312378119679 | 1122.99471398248 | 0.0 |
| | | 6 | | 77 | 2 | 4 | 12 | | 15 | |
| 6 | 68.0 | 29.4537851565220 | 11.0 | 16.7376321031480 | -10.0919183595317 | 56.2756081610393 | 9.04892310904004 | 12.7079254977041 | 824.909687560199 | 1.0 |
| | | 27 | | 48 | 29 | 44 | 1 | 88 | 4 | |
| 7 | 77.0 | 35.58792924 | 76.0 | 3.881 | 0.727558133 | 21.7863 | 8.12555 | 17.2615 | 618.272 | 0.0 |
| 8 | 48.0 | 28.125 | 90.0 | 2.54 | 0.56388 | 15.5325 | 10.22231 | 16.11032 | 1698.44 | 1.0 |
| 9 | 58.0 | 29.15451895 | 139.0 | 16.582 | 5.685415067 | 22.8884 | 10.26266 | 13.97399 | 923.886 | 1.0 |
| 10 | 68.0 | 21.36752137 | 77.0 | 3.226 | 0.612724933 | 9.8827 | 7.16956 | 12.766 | 928.22 | 0.0 |

## B.6 Audit Trace: Fold 6 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 52.0 | 22.2965620358864 | 99.0 | 6.27047343018537 | 0.92895772650270 | -3.25023382724732 | 7.85173921984758 | 40.6807992475795 | 930.011454753819 | 1.0 |
| | | 17 | | 4 | 69 | 63 | 6 | 2 | 3 | |
| 2 | 58.0 | 25.2853708157300 | 65.0 | 7.39536446345433 | 0.87383869960308 | 19.9741840379125 | 2.41950510745843 | 16.3414256597543 | 522.881503107451 | 1.0 |
| | | 83 | | 2 | 38 | 9 | 06 | 8 | 6 | |
| 3 | 48.0 | 19.8212038820515 | 81.0 | 3.85803483851392 | 0.61606063233596 | 14.3193455310801 | 13.5696590114439 | 9.37420496628787 | 156.926299797786 | 0.0 |
| | | 74 | | 76 | 71 | 31 | 39 | 9 | 95 | |
| 4 | 35.0 | 32.8176415673864 | 85.0 | 5.89411316031532 | 2.18287397833427 | 57.2684966043063 | 8.75455408454754 | 21.7130324662180 | 314.161607841338 | 0.0 |
| | | 8 | | 3 | 03 | | | 8 | 4 | |
| 5 | 72.0 | 34.1802304259379 | 100.0 | 7.63649202680516 | 1.33579564875040 | 17.8454928359766 | 10.8335920764879 | 10.9878078472853 | 410.231661622858 | 0.0 |
| | | 6 | | 8 | 81 | 85 | 67 | 28 | 3 | |
| 6 | 56.0 | 28.0531249457403 | 123.0 | 20.0440879333276 | 5.53209993821000 | 28.3488696267061 | 11.8434166425614 | 24.9714855006161 | 669.991200669441 | 1.0 |
| | | 26 | | 1 | 1 | 33 | 9 | 4 | 7 | |
| 7 | 74.0 | 28.65013774 | 88.0 | 3.012 | 0.6538048 | 31.1233 | 7.65222 | 18.35574 | 572.401 | 1.0 |
| 8 | 65.0 | 32.05 | 97.0 | 5.73 | 1.370998 | 61.48 | 22.54 | 10.33 | 314.05 | 1.0 |
| 9 | 76.0 | 27.2 | 94.0 | 14.07 | 3.262364 | 35.891 | 9.34663 | 8.4156 | 377.227 | 0.0 |
| 10 | 65.0 | 29.666548 | 85.0 | 14.649 | 3.071407 | 26.5166 | 7.28287 | 19.46324 | 1698.44 | 1.0 |

# TabuLLM: AI Benchmark Report

## B.7 Audit Trace: Fold 7 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 60.0 | 21.8405931179563 | 100.0 | 3.99779249166051 | 0.77295160151666 | 28.6594694445852 | 19.6174712871538 | 24.0439196619329 | 343.838674457172 | 1.0 |
| | | 2 | | | 79 | 4 | 72 | 96 | 8 | |
| 2 | 34.0 | 23.7575878416257 | 100.0 | 4.32389222478067 | 0.68286957446107 | 17.6625304949927 | 22.5882729470651 | 14.1828782386018 | 318.847315759505 | 1.0 |
| | | 7 | | 5 | 52 | 87 | 14 | 02 | 4 | |
| 3 | 71.0 | 27.5977406358896 | 131.0 | -3.95007005667256 | 2.20502158087009 | 28.2411548447654 | 8.06705986872774 | 11.9919079190553 | 253.010657459776 | 0.0 |
| | | 47 | | 73 | 7 | 15 | 8 | 47 | 44 | |
| 4 | 64.0 | 24.2770403360681 | 105.0 | 10.0859274504683 | 0.14375438234728 | 71.4337454524194 | 4.70900085832404 | 38.3208479655956 | 1156.69712148806 | 1.0 |
| | | 93 | | 61 | 8 | 7 | 9 | 96 | 12 | |
| 5 | 71.0 | 25.8095090473338 | 107.0 | 34.5944499602122 | -8.38114605270019 | 53.8351094807859 | 6.57979276136730 | -2.71454764781502 | 829.72143110612 | 0.0 |
| | | 6 | | 16 | | 74 | 5 | 1 | | |
| 6 | 95.0 | 20.4414363353070 | 93.0 | 9.46979813306044 | 6.18136144906161 | 8.11731153442223 | 8.76428484073404 | 8.97321517614732 | 872.232767448507 | 1.0 |
| | | 6 | | 7 | 2 | 4 | 6 | 8 | 5 | |
| 7 | 35.0 | 30.27681661 | 84.0 | 4.376 | 0.9067072 | 39.2134 | 9.048185 | 16.43706 | 733.797 | 0.0 |
| 8 | 69.0 | 32.5 | 93.0 | 5.43 | 1.245642 | 15.145 | 11.78796 | 11.78796 | 270.142 | 0.0 |
| 9 | 58.0 | 29.15451895 | 139.0 | 16.582 | 5.685415067 | 22.8884 | 10.26266 | 13.97399 | 923.886 | 1.0 |
| 10 | 49.0 | 29.77777778 | 70.0 | 8.396 | 1.449709333 | 51.3387 | 10.73174 | 20.76801 | 602.486 | 1.0 |

## B.8 Audit Trace: Fold 8 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 73.0 | 23.6113236826349 | 219.0 | 4.11019481934169 | 3.53832493424561 | 54.7577807521628 | 15.4241930848215 | 8.56639400146421 | 748.651708952837 | 1.0 |
| | | 32 | | 8 | 6 | 3 | 87 | 4 | | |
| 2 | 49.0 | 33.4104420005414 | 83.0 | 14.6958724188502 | 2.98095077339919 | 27.6577440616236 | 7.35196425282732 | 7.10337595064644 | 28.8317027228774 | 1.0 |
| | | 94 | | 96 | 27 | 53 | | | 8 | |
| 3 | 54.0 | 32.2471280419186 | 70.0 | 4.10130558294829 | 0.84290824171063 | 21.8331413114933 | 14.4619700306205 | 8.83776763301241 | 911.761625738062 | 0.0 |
| | | | | 9 | 1 | 75 | 84 | 5 | 5 | |
| 4 | 37.0 | 33.7239639698194 | 101.0 | 15.4079639752075 | 6.37009602867709 | 29.8772045178480 | 21.8548006353494 | 33.0652975197252 | 831.156842849001 | 1.0 |
| | | 65 | | 58 | 5 | 07 | 2 | | 7 | |
| 5 | 50.0 | 29.2457082303875 | 162.0 | 43.0792797709221 | 6.55793259142674 | 40.8336380579760 | 14.5245533993930 | 4.31275552754524 | 1157.89855782061 | 1.0 |
| | | 03 | | 9 | 15 | 46 | 03 | 4 | 25 | |
| 6 | 44.0 | 34.4517207661558 | 87.0 | 21.2214636394981 | 0.17785672660620 | 24.2832882170831 | 5.90151028821561 | 22.4971851880948 | 19.3331977643756 | 0.0 |
| | | | | 6 | 41 | 1 | 1 | 32 | 55 | |
| 7 | 45.0 | 37.03560819 | 83.0 | 6.76 | 1.383997333 | 39.9802 | 4.617125 | 8.70448 | 586.173 | 0.0 |
| 8 | 61.0 | 32.03895937 | 85.0 | 18.077 | 3.790144333 | 30.7729 | 7.780255 | 13.68392 | 444.395 | 0.0 |
| 9 | 46.0 | 22.21 | 86.0 | 36.94 | 7.836205333 | 10.16 | 9.76 | 5.68 | 312.0 | 1.0 |
| 10 | 85.0 | 27.68877813 | 196.0 | 51.814 | 25.05034187 | 70.8824 | 7.901685 | 55.2153 | 1078.359 | 1.0 |

# TabuLLM: AI Benchmark Report

## B.9 Audit Trace: Fold 9 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 44.0 | 34.4654468402156 | 82.0 | 14.5019680027806 | 1.14578723583021 | 50.1932745895778 | 8.56002762640291 | 33.1236907008861 | 1277.74294176010 | 0.0 |
| | | 7 | | 93 | 06 | 9 | | 6 | 52 | |
| 2 | 62.0 | 29.0266067285142 | 88.0 | 13.2809960179669 | 1.59320477755018 | 40.6518034622985 | 4.66308318376958 | 12.6981610042834 | 366.506900947705 | 0.0 |
| | | 56 | | 65 | 56 | 6 | 8 | 84 | 3 | |
| 3 | 35.0 | 29.6137098408057 | 85.0 | 4.38519706216578 | 0.66357295066289 | 37.2626886946759 | 20.5283132942039 | 18.0893529624884 | 39.1650379479517 | 1.0 |
| | | 97 | | 5 | 83 | 94 | 3 | 94 | 6 | |
| 4 | 41.0 | 27.7933226970325 | 80.0 | 4.41404942884870 | 0.60044491959853 | 21.8823046704709 | 13.6761479867172 | 10.8385787836777 | 897.147713479638 | 1.0 |
| | | 04 | | 3 | 62 | 1 | 04 | 33 | 2 | |
| 5 | 74.0 | 31.9901207572199 | 137.0 | 5.47900830626909 | 3.64402772025197 | 7.21371807605357 | 5.01116651682136 | 9.03931245856263 | 272.708740732937 | 1.0 |
| | | 98 | | 6 | 9 | 8 | 3 | 7 | 03 | |
| 6 | 78.0 | 25.9406666816343 | 81.0 | 4.92725018685009 | 1.17454516063531 | 15.5602657394733 | 13.1196596468882 | 7.32723811363238 | 141.460162290955 | 1.0 |
| | | 47 | | 4 | 28 | 87 | 46 | 65 | 34 | |
| 7 | 78.0 | 25.3 | 60.0 | 3.508 | 0.519184 | 6.633 | 10.567295 | 4.6638 | 209.749 | 0.0 |
| 8 | 52.0 | 30.8012487 | 87.0 | 30.212 | 6.4834952 | 29.2739 | 6.26854 | 24.24591 | 764.667 | 1.0 |
| 9 | 45.0 | 20.26 | 92.0 | 3.44 | 0.780650667 | 7.65 | 16.67 | 7.84 | 193.87 | 1.0 |
| 10 | 59.0 | 28.67262608 | 77.0 | 3.188 | 0.605507467 | 17.022 | 16.44048 | 31.6904 | 910.489 | 1.0 |

## B.10 Audit Trace: Fold 10 | Gen: 100%

| # | Age | BMI | Glucose | Insulin | HOMA | Leptin | Adiponectin | Resistin | MCP.1 | Classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 67.0 | 24.3873918873009 | 104.0 | 9.98240996992934 | 2.08652658152073 | 57.6607617957587 | 5.92620160891947 | 0.23839487975883 | 826.613037227517 | 1.0 |
| | | 86 | | | 83 | 24 | 3 | 6 | 5 | |
| 2 | 58.0 | 30.7377624985863 | 96.0 | 28.6478170141663 | 2.43692497291477 | 33.1600518582730 | 6.74622804638839 | 22.3059425548644 | 472.800840841914 | 1.0 |
| | | 88 | | 04 | 57 | 2 | 75 | 2 | 46 | |
| 3 | 93.0 | 28.2300467678815 | 87.0 | 3.92399350755154 | 0.91502230645256 | 13.9207053554592 | 3.76367871084859 | 18.0972321935704 | 586.301564703963 | 0.0 |
| | | 96 | | 1 | 41 | 48 | 03 | 02 | 5 | |
| 4 | 21.0 | 37.1458426389830 | 97.0 | 15.0121543828985 | 2.19994690858938 | 27.1724281818446 | 24.0715009798584 | 27.4103640247526 | 447.445521750288 | 0.0 |
| | | 7 | | 95 | 9 | 48 | 03 | 53 | 04 | |
| 5 | 57.0 | 30.0261149260929 | 96.0 | 10.5687791267156 | 2.63249674849088 | 60.9587278278030 | 12.9358939254315 | 12.3088549358902 | 379.408764175092 | 0.0 |
| | | 44 | | 77 | 6 | 45 | 64 | 8 | 6 | |
| 6 | 58.0 | 33.4399672796849 | 72.0 | 16.9981760669270 | 3.92858793153492 | 1.70061924605688 | 9.11462950969622 | 33.5504254693104 | 740.157857762048 | 0.0 |
| | | 64 | | 76 | 72 | 83 | 7 | 54 | | |
| 7 | 64.0 | 34.5297228 | 95.0 | 4.427 | 1.037393667 | 21.2117 | 5.46262 | 6.70188 | 252.449 | 0.0 |
| 8 | 32.0 | 36.51263743 | 87.0 | 14.026 | 3.0099796 | 49.3727 | 5.1 | 17.10223 | 588.46 | 0.0 |
| 9 | 43.0 | 31.25 | 103.0 | 4.328 | 1.099600533 | 25.7816 | 12.71896 | 38.6531 | 775.322 | 1.0 |
| 10 | 57.0 | 34.83814777 | 95.0 | 12.548 | 2.940414667 | 33.1612 | 2.36495 | 9.9542 | 655.834 | 1.0 |